

Question no#1:

```
#include <iostream>
```

```
#include <string>
```

```
#include <exception>
```

```
using namespace std;
```

```
class InvalidPasswordException : public exception {
```

```
public:
```

```
    const char* what() const throw() {
```

```
        return "Exception: Password must be at least 6 characters and contain a digit.";
```

```
    }
```

```
};
```

```
void validatePassword(string password) {
```

```
    if (password.length() < 6) {
```

```
        throw InvalidPasswordException();
```

```
    }
```

```
    bool hasDigit = false;
```

```
    for (char c : password) {
```

```
        if (isdigit(c)) {
```

```
            hasDigit = true;
```

```
            break;
```

```
        }
```

```
    }
```

```
    if (!hasDigit) {
```

```
        throw InvalidPasswordException();
    }
}

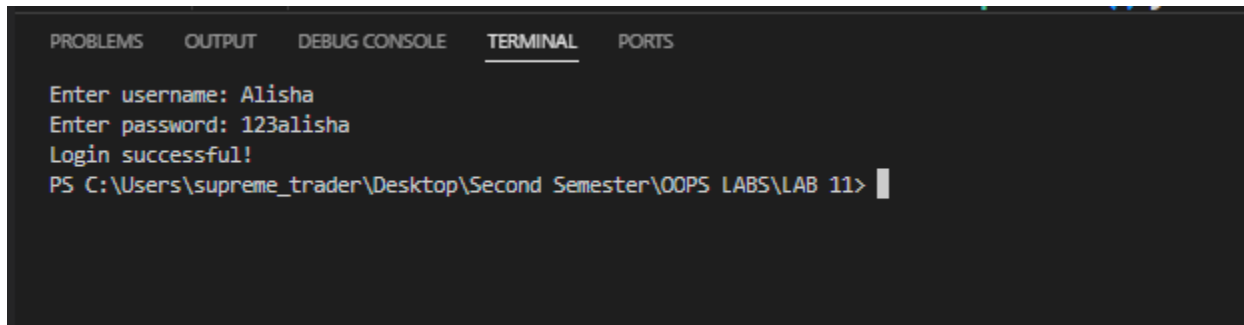
int main() {
    string username, password;

    cout << "Enter username: ";
    cin >> username;
    cout << "Enter password: ";
    cin >> password;

    try {
        validatePassword(password);
        cout << "Login successful!" << endl;
    } catch (const InvalidPasswordException& e) {
        cout << e.what() << endl;
    }

    return 0;
}
```

Output:

A screenshot of a terminal window with a dark background. At the top, there are tabs labeled 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is selected and underlined), and 'PORTS'. The terminal shows the following text: 'Enter username: Alisha', 'Enter password: 123alisha', 'Login successful!', and 'PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 11>'.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Enter username: Alisha
Enter password: 123alisha
Login successful!
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 11>
```

Question no#2:

```
#include <iostream>
```

```
#include <vector>
```

```
#include <string>
```

```
using namespace std;
```

```
class Stack {
```

```
private:
```

```
    vector<char> elements;
```

```
public:
```

```
    void push(char c) {
```

```
        elements.push_back(c);
```

```
    }
```

```
    void pop() {
```

```
    if (!elements.empty()) {  
        elements.pop_back();  
    }  
}
```

```
char peek() {  
    if (!elements.empty()) {  
        return elements.back();  
    }  
    return '\0';  
}
```

```
bool isEmpty() {  
    return elements.empty();  
}
```

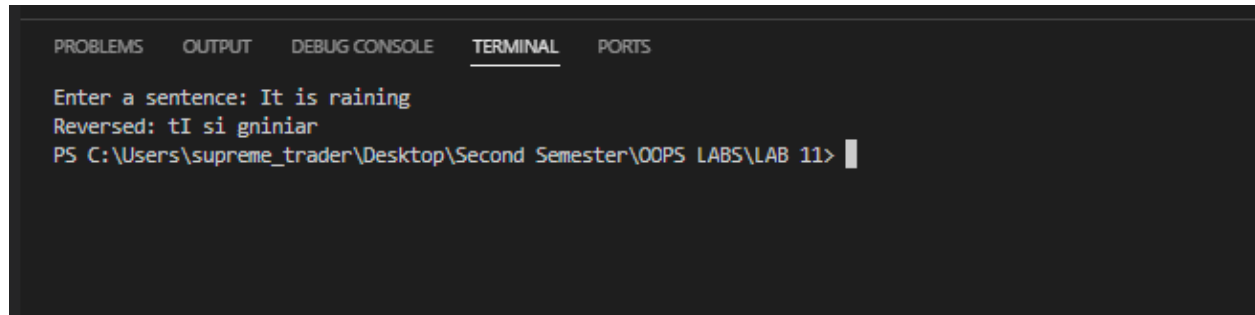
```
string reverse(string sentence) {  
    string result = "";  
    for (int i = 0; i <= sentence.length(); i++) {  
        if (i == sentence.length() || sentence[i] == ' ') {  
            while (!isEmpty()) {  
                result += peek();  
                pop();  
            }  
        }  
    }  
}
```

```
    }  
    if (i < sentence.length()) {  
        result += " ";  
    }  
} else {  
    push(sentence[i]);  
}  
}  
return result;  
}  
};
```

```
int main() {  
    Stack s;  
    string sentence;  
  
    cout << "Enter a sentence: ";  
    getline(cin, sentence);  
  
    string reversedSentence = s.reverse(sentence);  
    cout << "Reversed: " << reversedSentence << endl;  
  
    return 0;
```

```
}
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Enter a sentence: It is raining
Reversed: tI si gninar
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 11>
```

Question no#3:

```
#include <iostream>
```

```
#include <vector>
```

```
#include <algorithm>
```

```
#include <ctime>
```

```
using namespace std;
```

```
void customSort(vector<int>& arr) {
```

```
    int n = arr.size();
```

```
    for (int i = 0; i < n - 1; i++) {
```

```
        for (int j = 0; j < n - i - 1; j++) {
```

```
            if (arr[j] > arr[j + 1]) {
```

```
                swap(arr[j], arr[j + 1]);
```

```
            }
```

```
        }
```

```
    }
```

```
}
```

```
int main() {
```

```
    int n;
```

```
    cout << "Enter number of integers: ";
```

```
    cin >> n;
```

```
    vector<int> vec1(n);
```

```
    vector<int> vec2(n);
```

```
    cout << "Enter the integers: ";
```

```
    for (int i = 0; i < n; i++) {
```

```
        cin >> vec1[i];
```

```
        vec2[i] = vec1[i];
```

```
    }
```

```
    clock_t start1 = clock();
```

```
    customSort(vec1);
```

```
    clock_t end1 = clock();
```

```
    clock_t start2 = clock();
```

```
    sort(vec2.begin(), vec2.end());
```

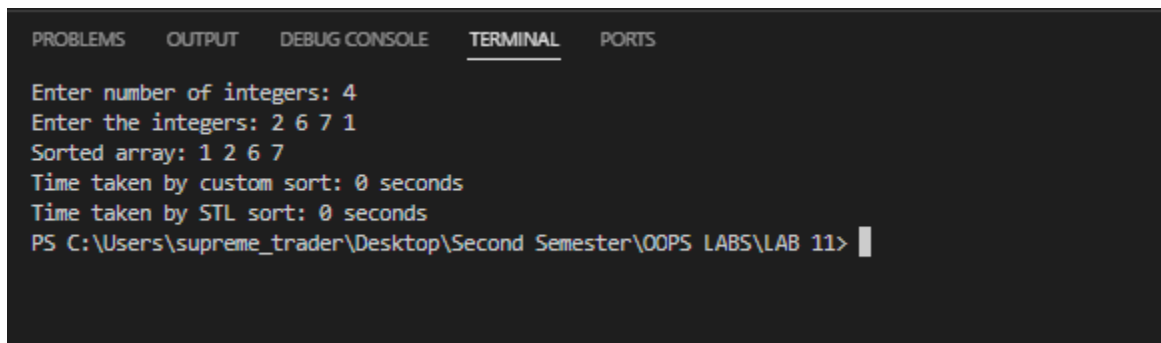
```
    clock_t end2 = clock();
```

```
    cout << "Sorted array: ";
```

```
    for (int i = 0; i < n; i++) {
```

```
        cout << vec1[i] << " ";  
    }  
    cout << endl;  
  
    double time1 = double(end1 - start1) / CLOCKS_PER_SEC;  
    double time2 = double(end2 - start2) / CLOCKS_PER_SEC;  
  
    cout << "Time taken by custom sort: " << time1 << " seconds" << endl;  
    cout << "Time taken by STL sort: " << time2 << " seconds" << endl;  
  
    return 0;  
}
```

Output:



The screenshot shows a terminal window with the following output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
Enter number of integers: 4  
Enter the integers: 2 6 7 1  
Sorted array: 1 2 6 7  
Time taken by custom sort: 0 seconds  
Time taken by STL sort: 0 seconds  
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 11>
```


Question no#4:

```
#include <iostream>
```

```
#include <map>
```

```
#include <string>
```

```
using namespace std;
```

```
int main() {
```

```
    map<string, char> studentGrades;
```

```
    int choice;
```

```
    string name;
```

```
    char grade;
```

```
    while (true) {
```

```
        cout << "\n1. Add Student\n2. Retrieve Grade\n3. Update Grade\n4. Delete Student\n5.  
Display All\n6. Exit\nChoice: ";
```

```
        cin >> choice;
```

```
        if (choice == 1) {
```

```
            cout << "Enter name: ";
```

```
            cin >> name;
```

```
            cout << "Enter grade: ";
```

```
            cin >> grade;
```

```
            studentGrades[name] = grade;
```

```
        }
```

```
        else if (choice == 2) {
```

```
    cout << "Enter name: ";
    cin >> name;
    if (studentGrades.find(name) != studentGrades.end()) {
        cout << "Grade: " << studentGrades[name] << endl;
    } else {
        cout << "Student not found." << endl;
    }
}

else if (choice == 3) {
    cout << "Enter name: ";
    cin >> name;
    if (studentGrades.find(name) != studentGrades.end()) {
        cout << "Enter new grade: ";
        cin >> grade;
        studentGrades[name] = grade;
    } else {
        cout << "Student not found." << endl;
    }
}

else if (choice == 4) {
    cout << "Enter name: ";
    cin >> name;
    if (studentGrades.erase(name)) {
        cout << "Student deleted." << endl;
    } else {
        cout << "Student not found." << endl;
    }
}
```

```
    }  
}  
else if (choice == 5) {  
    cout << "Student Grades:" << endl;  
    for (auto const& pair : studentGrades) {  
        cout << pair.first << ": " << pair.second << endl;  
    }  
}  
else if (choice == 6) {  
    break;  
}  
else {  
    cout << "Invalid choice." << endl;  
}  
}  
  
return 0;  
}
```

Output:

```
1. Add Student
2. Retrieve Grade
3. Update Grade
4. Delete Student
5. Display All
6. Exit
```

```
Choice: 1
Enter name: Alisha
Enter grade: A
```

```
1. Add Student
2. Retrieve Grade
3. Update Grade
4. Delete Student
5. Display All
6. Exit
```

```
Choice: 2
Enter name: Alisha
Grade: A
```

```
1. Add Student
2. Retrieve Grade
3. Update Grade
4. Delete Student
5. Display All
6. Exit
```

```
Choice: 3
Enter name: Alisha
Enter new grade: B
```

```
1. Add Student
2. Retrieve Grade
3. Update Grade
4. Delete Student
5. Display All
6. Exit
```

```
Choice: 2
Enter name: Sara
Student not found.
```

```
1. Add Student
2. Retrieve Grade
3. Update Grade
4. Delete Student
5. Display All
6. Exit
```

```
Choice: 2
Enter name: Alisha
Grade: B
```

```
1. Add Student
2. Retrieve Grade
3. Update Grade
4. Delete Student
5. Display All
6. Exit
```

```
Choice: 4
Enter name: Alisha
Student deleted.
```

```
1. Add Student
2. Retrieve Grade
3. Update Grade
4. Delete Student
5. Display All
6. Exit
```

```
Choice: 5
Student Grades:
```

```
1. Add Student
```

Question no#5:

```
#include <iostream>
```

```
#include <set>
```

```
#include <string>
```

```
using namespace std;
```

```
int main() {
```

```
    set<string> guests;
```

```
    string name;
```

```
    int choice;
```

```
    while (true) {
```

```
        cout << "\n1. Add Guest\n2. Display Guests\n3. Count Guests\n4. Exit\nChoice: ";
```

```
        cin >> choice;
```

```
        if (choice == 1) {
```

```
            cout << "Enter guest name: ";
```

```
            cin >> name;
```

```
            guests.insert(name);
```

```
            cout << "Guest added." << endl;
```

```
        }
```

```
    else if (choice == 2) {  
        cout << "Guest List:" << endl;  
        for (const string& guest : guests) {  
            cout << guest << endl;  
        }  
    }  
    else if (choice == 3) {  
        cout << "Total unique guests: " << guests.size() << endl;  
    }  
    else if (choice == 4) {  
        break;  
    }  
    else {  
        cout << "Invalid choice." << endl;  
    }  
}  
  
return 0;  
}
```

Output:

```
1. Add Guest
2. Display Guests
3. Count Guests
4. Exit
Choice: 1
Enter guest name: Alisha
Guest added.

1. Add Guest
2. Display Guests
3. Count Guests
4. Exit
Choice: 2
Guest List:
Alisha

1. Add Guest
2. Display Guests
3. Count Guests
4. Exit
Choice: 1
Enter guest name: Sara
Guest added.

1. Add Guest
2. Display Guests
3. Count Guests
4. Exit
Choice: 2
Guest List:
Alisha
Sara

1. Add Guest
2. Display Guests
3. Count Guests
4. Exit
Choice: 3
Total unique guests: 2

1. Add Guest
2. Display Guests
3. Count Guests
4. Exit
Choice: 4
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 11> 
```